Dynamic Macroeconomics II

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Topic 1: Models of Real Business Cycles

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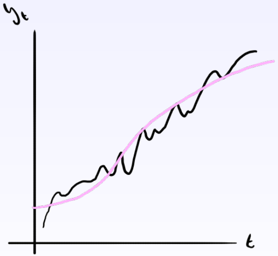
- References: Cooley's textbook (1995), Prescott (1986).
- Definition: Lucas (1977): fluctuations of output around a trend, and comovement of other economics series with output.
- How to distinguish cycle and trend?
- Neoclassical theory uses **same** theory to explain growth and cycles
- After measurement, we want a model to understand sources of the cycle
- A classical reference is Kydland & Prescott (1982).

We use the Hodrick-Prescott (HP) filter to detrend data and observe cyclical components.

- We can break a time series Y_t down into two components: trend and cycle.



$$Y_t = Y_t^c + Y_t^g$$

corr $(y_t^c, C_t^c) > 0$

- We obtain those components by solving the following problem (the Hodrick-Prescott filter):

min error c.r datos   min variación suavidad

$$\min_{Y_t^g} \quad \sum_{t=1}^{T}(Y_t - Y_t^g)^2 + \lambda \sum_{t=2}^{T-1}[(Y_{t+1}^g - Y_t^g) - (Y_t^g - Y_{t-1}^g)]^2$$

tend se debe parecer a los datos

where $\lambda \geq 0$ is a smoothing parameter.

- The first part $(Y_{t+1} - Y_t^g)$ is the cyclical component and the second one $(Y_{t+1}^g - Y_t^g) - (Y_t^g - Y_{t-1}^g)$ is the difference in changes in the trend.
- λ acts as a penalty for large cyclical components.
- If $\lambda = 0$ then, $Y_t = Y_t^g$.
- On the other hand, if $\lambda \to \infty$ the trend tends to be linear.
- The higher the value of λ, the smoother the trend is.
- You can do this in Eviews, Matlab, etc.

Con estos λ y calculamos la tend. se elimina el componente cíclico que dura

- In practice, we use $\lambda = 1600$ for quarterly data and $\lambda = 100$ for annual data.
- The reason is historical (see Cooley and Prescott 1995). When calculating their **trend**, Hodrick and Prescott's goal was to eliminate fluctuations at frequencies higher? lower than 8 years.
- Economists think of cycles ocurring with a frequency of 3-5 years. That is what Burns and Mitchell (1946) characterized as usual business cycle frequency in the United States.

$\ln\left(X_{t+1}/X_t\right) \simeq$ tasa crecimiento // $X_{t+1}/X_t \equiv$ factor crec

$100\left(X_{t+1}/X_t - 1\right) \equiv$ tasa crec

- It is useful to work with logarithms. Recall that $\log\left(\frac{x_{t+1}}{x_t}\right)$ approximates the growth rate of a variable x.
- Then

cociente

(qué tan lejos está la tendencia de los datos)

y_t^c, C_t^c v.a.

↳ estacionarias

$$\underbrace{\log\left(\frac{Y_t}{Y_t^g}\right)}_{\text{distancia en pp, i.e., parte cíclica}} = \log\left(Y_t\right) - \log\left(Y_t^g\right) \approx \log\left(Y_t^c\right)$$

↳ Variables estacionarias es interesante calcular sus momentos

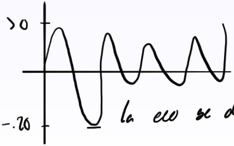
- Interpretation: If we multiply by 100, get percentage difference from trend.
- We measure the volatility of (the cyclical component of) a variable x by looking at its standard deviation σ_x.

$\sigma^c/\sigma_y > 1?$

< 1 suav consumo

$= 0$ suav perf.

- We can also calculate correlations with output



la eco se desvió 20 pts porc de su tendencia

Interpretación a las magnitudes del ciclo eco

Definitions:

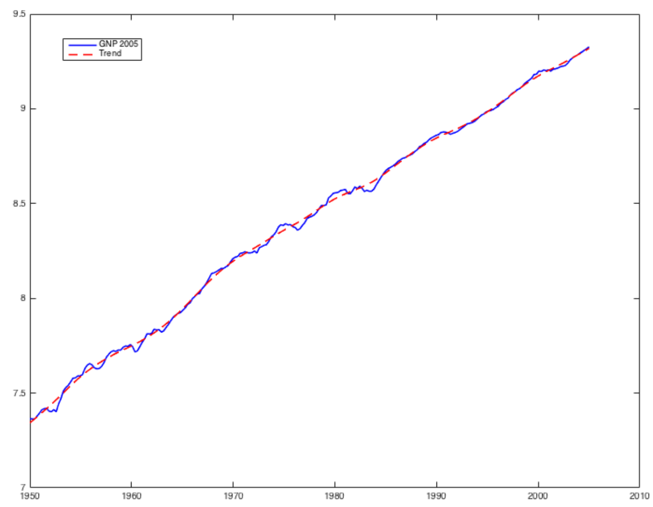
- A variable is **procyclical** if its correlation with GDP is greater than zero. $Corr(c, y) > 0$
- A variable is **countercyclical** if its correlation with GDP is negative. $Corr(NX, y) < 0$ x crisis
- Also, if the correlation of a variable with GDP is around zero, that variable is called **acyclical**.
- We look at how strongly some variables are correlated with GDP led or lagged a number of periods.
- By observing that, we can determine which variables **lead** or **lag** the cycle.
- Example: **past** values of Money are highly positively correlated with ouptut. We say Money **leads** the cycle

1. Hours are about as a volatile as output: the business cycle is manifested in the labor market (Cooley and Prescott 1995).
2. Employment (number of employees) varies almost as much as total hours worked.
3. Average hours worked (by employee) vary much less than total hours worked.
4. This suggests that extensive margin (number of workers) is more variable than intensive margin (hours worked).
5. Consumption is less volatile than GDP.
6. Investment is more volatile than GDP.
7. Labor productivity (average product of labor) is negatively correlated or uncorrelated with hours worked.

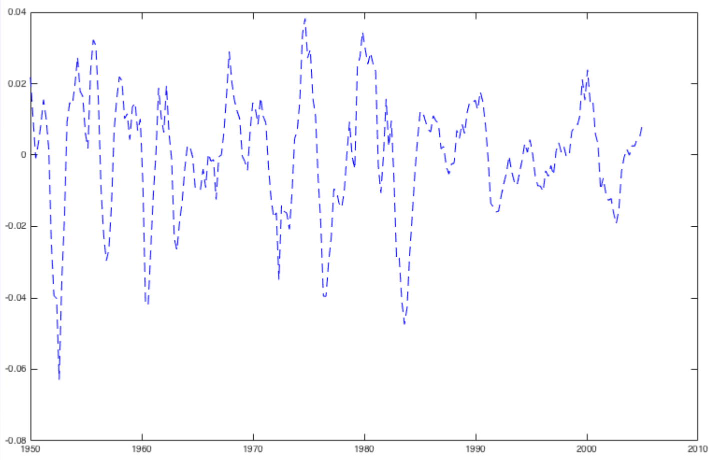
$Corr(horas, y/l) < 0$

Business Cycles Stylized Facts: Mexico

1. Consumption is more volatile than GDP.
2. Cyclical component of employment relative to the one of GDP is lower in Mexico than in Canada (Fernandez and Meza 2015).
3. Correlation between employment and GDP is smaller in Mexico than in Canada (Fernandez and Meza 2015).



Log of real GNP and its trend component for US, 1947-2005.



HP filtered cyclical component for US, 1947-2005.

- RBC methodology emphasizes building a model, simulating it and comparing model's predictions with data.
- Kydland & Prescott (1982) studied the predictions of a neoclassical growth model with stochastic shocks to aggregate productivity.
- The model fails to explain the volatility of hours worked.
- Correlation of labor productivity (average product of labor) with hours worked is near one, when in data is close to zero.
- The methodology of this paper has been widely used to study many different sources of real business cycles.
- Some extensions are Hansen (1985), who proposes indivisible labor, and McGrattan (1994), who works with fiscal shocks.

McGrattan (1994): Standard model

- Consumer maximizes expected utility subject to budget constrain, law of motion of capital, and time constraint, given initial conditions x_0:

$$\max \quad E_0 \sum_{t=0}^{\infty} \beta^t [u(c_t, \ell_t) \mid x_0]$$

s.t.

$$c_t + i_t \leq r_t k_t + w_t n_t \tag{1}$$

$$i_t = k_{t+1} - (1-\delta)k_t \tag{2}$$

$$1 = l_t + n_t \tag{3}$$

- l_t is leisure, n_t is labor. $0 < \beta < 1, 0 < \delta < 1$
- Recall the difference between net investment $(k_{t+1} - k_t)$ and gross investment $(k_{t+1} - k_t + \delta k_t)$.

- Representative firm solves:

$$\max \quad y_t - w_t n_t - r_t k_t$$

s.t.

$$y_t = \lambda_t f(k_t, n_t) \tag{4}$$

- λ_t is the total factor productivity (TFP).
- The consumer does all the investment in the economy, so the firm's problem is static.
- Solving the firm's problem, we obtain:

$$\begin{aligned} w_t &= \lambda_t f_{n_t} = MP_{n_t} \\ r_t &= \lambda_t f_{k_t} = MP_{k_t} \end{aligned}$$

- Since this is a competitive economy, agents take prices as given, as well as the initial level of capital k_0.
- Another component of the model: Feasibility :

$$c_t + i_t = y_t$$

- A **state vector** x_t summarizes the information about the current position of a system that is relevant for determining its future. For this economy, $x_t = \{k_t, \lambda_t\}$.
- We assume TFP is stochastic and follows a first-order autoregressive model AR(1):

$$\lambda_{t+1} = (1-\rho_\lambda)\bar{\lambda} + \rho_\lambda \lambda_t + \varepsilon_{t+1}, \quad \varepsilon \sim (0, \sigma_\varepsilon^2)$$

where $-1 < \rho_\lambda < 1$ represents the persistence of the shock.

- Note that in the consumer's problem I am **not** describing carefully the uncertainty.
- Solution must be valid for all periods t and all states of the world.
- The solution to this problem is a function $k_{t+1} = k(k_t, \lambda_t)$
- Which provides us with functions $c_t = c(k_t, \lambda_t), i_t = i(k_t, \lambda_t), n_t = n(k_t, \lambda_t)$.

- To solve the competitive equilibrium, we need to choose functional forms for utility and technology. Then, let:

$$u(c_t, \ell_t) = \ln(c_t) + \gamma \ln(\ell_t)$$
$$f(k_t, n_t) = k_t^{\theta} n_t^{1-\theta}$$

- $\gamma > 0, 0 < \theta < 1$

Calibration: Definition

- Find parameter values such that the model replicates some features of the economy.
- Do NOT calibrate to reproduce exactly what we aim to explain.
- We would be answering main question by construction, model has no explanatory power

Calibration: Implementation

- Obtain

$$\lambda_t = \frac{y_t}{k_t^{\theta} n_t^{1-\theta}}$$

- If we have measured $lambda_t$, in the AR(1) equation for productivity normalize $\bar{\lambda}$ to 1. This parameter only affects levels, not cycles. Use econometric techniques to estimate the persistence ρ_λ.
- Since time series for capital are generally not available, we need to construct it using aggregate investment data from national income accounts and the perpetual inventory methods

Medir λ_t

Quitar tendencia de crecimiento a λ_t

Recall

$\ln(1+g) = g$

- In real world, TFP growths along a trend, which is incompatible with an autoregressive formulation.
- One way to deal with this problem is proposed by McGrattan: Assume TFP growths geometrically, $\lambda_t = \lambda_0(1+g)^t$, and regress $\ln(TFP_t) = \ln(\lambda_0) + t\ln(1+g) \approx \ln(\lambda_0) + gt$.
- Regression: $\ln TFP_t = \beta_0 + \beta_1 t$
- $\ln \lambda_t$ consistent with model is equal to $\ln TFP_t - \beta_1 t$.
- Then, by removing the trend, we can estimate the AR(1) process and obtain values for ρ_λ and σ_ε.

- Given data on real interest rate and using the steady state conditions, calibrate β as:

$$(1+i) = (1+r)(1+\pi) \qquad \beta = \frac{1}{1+r} \Rightarrow \text{Euler}\Big|_{\text{ss no estocastico}} \to \beta$$

r Datos reales

- For an annual REAL interest rate of 4 %, we obtain $\beta = 0.96$, while for quarterly data it is common to get a value of $\beta = 0.99$.
- What amount of time does t represent? A month, a quarter, a year?
- Value of β depends on our choice.

- To calibrate δ, use the capital stock and the law of motion of capital. For the US there are data, for Mexico not (need to check recent measurements by INEGI).
- Regression based on $i_t - (k_{t+1} - k_t) = \delta k_t$
- McGrattan finds quarterly value of $\delta = 0.023$.
- This implies that annual depreciation rate equals $(1 - 0.023)^4 \approx 0.91 = 1 - \delta_{annual}$. Then, $\delta_{annual} = 0.09$.

Calibrating θ

- Steady state Euler equation:

$$1 = \beta\left[\theta\frac{y}{k} + (1-\delta)\right]$$

- Have to be careful when looking at ratio K/Y from the data, since K is a stock and Y is a flow.
- Quarterly data gives a value of K/Y close to 11, while using annual data $K/Y \approx 3$ in developed countries.
- Then, it is very common to get values of θ close to $1/3$.

$wn/y = 1-\theta \Rightarrow w = (1-\theta)y/n$

Calibrating γ

- The utility function is $u(c_t, n_t) = \ln(c_t) + \gamma \ln(1 - n_t)$, where γ is the preference for leisure.
- For US economy, $n \approx 0.27$.
- Then, we can calibrate γ using the steady state consumption-leisure equation:

$$\frac{\gamma c}{1-n} = (1-\theta)\frac{y}{n}$$

- With the values we have calculated we get $\gamma \approx 2$.

- Calibrate the model and find a solution method (e.g. value function iteration, log-linearization plus undetermineed coefficients).
- Generate a random sequence $\{\lambda_t\}$.
- Simulate the model for a long number of periods T.
- Obtain the equilibrium sequences $\{c_t, y_t, n_t, i_t, k_{t+1}\}_t^T$.
- Apply the Hodrick-Prescott filter to the resulting sequences and find theoretical moments.
- Compare the model's statistics with data.
- Let's see Table 1 in McGrattan (1994).

U.S. Data vs. The Standard Kydland-Prescott Model and Hansen's Extension

Statistic	U.S. Data* 1947:1–1987:4	Model Standard†	Model Hansen**
Standard Deviation of			
Output	1.81	1.83 (.20)	2.27 (.24)
Consumption	.91	.67 (.09)	.77 (.10)
Investment	5.11	5.69 (.72)	7.33 (.96)
Capital Stock	.45	.45 (.08)	.57 (.11)
Hours Worked	1.52	.89 (.09)	1.57 (.16)
Productivity Pme	1.32	.97 (.11)	.78 (.10)
Correlation Between Hours Worked and Productivity	–.195	.94 (.01)	.84 (.03)

Source: McGrattan (1994).

Hansen (1985)

- Consumers either work a fixed number of hours N or not at all (0).
- Discrete choice.
- Choice set is not convex, so Hansen convexifies by using lotteries.
- That is, probabilities over 0 or N.
- Consumer chooses over lotteries.
- These assumptions imply a linear-in-leisure utility function: $u(c_t, \ell_t) = \ln(c_t) + \gamma_2 \ell_t$.
- See Table 1 in McGrattan (1994).

- Adds fiscal shocks to standard business cycles model.
- Taxes distort agents' behavior.
- The government taxes labor and capital income and runs a balanced budget (there is no public debt).
- Public expenditures enter into consumers' utility: $u(c_t + \pi g_t, \ell_t)$, where $\pi \geq 0$ is the preference for government expenditures.
- To balance the model, we introduce (endogenous) lump-sum transfers T_t: $g_t + T_t = \tau_{k_t}(r_t - \delta)k_t + \tau_{n_t} w_t n_t$, where g_t is exogenous public expenditures.
- Consumer's budget constraint: $c_t + i_t \leq (1 - \tau_{n_t})w_t n_t + r_t k_t - \tau_{k_t}(r_t - \delta)k_t + T_t$.

- Taxes follow a stochastic process which is known by consumers.
- Exogenous variables for this economy: $v_t = (\lambda_t, g_t, \tau_{k_t}, \tau_{n_t})$.
- Let $\bar{v}$ be the long-run value of v_t.
- Vector autoregression (VAR):

$$v_{t+1} = (I - \rho_v)\bar{v} + \rho_v v_t + \varepsilon_{t+1} \quad \varepsilon \sim (0, \Sigma)$$

  Same idea as an AR(1), but in vectorial form
- State vector: $x_t = (k_t, v_t)$.
- New feasibility condition: $c_t + i_t + g_t = y_t$.

- McGrattan estimates a value of $\pi \approx 0$. When measuring output McGrattan **added** government expenditures
- In the standard model $K/Y \approx 11$. Now, $K/Y \approx 8.3$.
- New values for $\theta = 0.359$ and $\gamma = 2.33$. Before $\theta = 0.355$ and $\gamma = 2.36$.

- Use the vector autoregression to calculate ρ_λ:

$$v_{t+1} = \begin{bmatrix} \lambda_t \\ g_t \\ \tau_{n_t} \\ \tau_{k_t} \end{bmatrix} \qquad \rho_v = \begin{bmatrix} \rho_\lambda & 0 & 0 & 0 \\ 0 & \rho_g & 0 & 0 \\ 0 & 0 & \rho_n & 0 \\ 0 & 0 & 0 & \rho_k \end{bmatrix}$$

- With these assumptions McGrattan estimates line by line.
- Variables in v_t have trends. Remove trends as before.
- Set $\bar{\lambda} = 1$ as before. Set $\bar{g}$ so that $\frac{\bar{g}}{\bar{y}} = 0.22$, $\bar{\tau}_k = 0.5$, $\bar{\tau}_n = 0.23$. Those are approximately the long run values of the shocks in the data.
- To estimate Σ, use VAR residuals.

- Let's see Table 2.
- Relative to Kydland & Prescott, the model does a good job in explaining the contemporaneous correlation between hours worked and productivity.
- This correlation decreases but is still positive, while in data is negative.
- Putting together indivisible labor and fiscal policy produces a negative correlation, smaller than in data.
- Fiscal shocks are important drivers for movements in hours and labor productivity.

Statistic	U.S. Data* 1947:1–1987:4	Model: Divisible Labor†	Model: Indivisible Labor**
Standard Deviation of			
Output	1.81	1.83 (.21)	2.23 (.26)
Consumption	.91	.98 (.12)	1.03 (.13)
Investment	5.11	5.53 (.77)	7.24 (1.10)
Capital Stock	.45	.44 (.09)	.57 (.12)
Hours Worked	1.52	1.31 (.15)	2.03 (.23)
Productivity	1.32	1.11 (.13)	1.10 (.14)
Correlation Between Hours Worked and Productivity	−.195	.14 (.13)	−.08 (.12)

Source: McGrattan (1994).

Intuition for results

- Consider a static economy in which consumer solves:

$$\max \quad u = \ln(c) + b \ln(1 - n)$$

s.t.

$$c \leq wn$$

- Solving the consumer's problem, we obtain an expression for labor supply that only depends on preferences:

$$n = \frac{1}{1 + b}$$

- Two-period model:

$$\max \quad u = \ln(c_1) + b\ln(1-n_1) + \beta[\ln(c_2) + b\ln(1-n_2)]$$

s.t.

$$c_1 + \frac{c_2}{1+r} = w_1 n_1 + \frac{w_2 n_2}{1+r}$$

- First order condition:

Si $1/\beta(1+r)\; w_2/w_1 \uparrow$ ent $n_1 \downarrow$

$$\frac{1-n_1}{1-n_2} = \frac{1}{\beta(1+r)}\frac{w_2}{w_1}$$

- Today's labor supply depends on current and future productivities, via real wages.
- A higher wage in period 1 relative to period 1 implies $n_1 > n_2$.
- When the interest rate r is high, agents work more on period 1. Consumer wants to transfer more resources from period 1 to 2 because incentive to save is larger.

- Adding taxes to the model gives us the following modified first order condition:

$$\frac{1-n_1}{1-n_2} = \frac{1}{\beta(1+r)} \frac{w_2}{w_1} \frac{1-\tau_2}{1-\tau_1}$$

- Last term on right-hand side vanishes if tax rates are equal across time.
- A higher tax today implies **less** labor today compared to tomorrow $n_1 < n_2$.
- Taxes on labor income are crucial to generate a small correlation between hours worked and labor productivity.

Exercise: What correlation between taxes and exogenous productivity is needed to produce low correlation between hours worked and labor productivity?